SEMICONDUCTOR MISSION AND ELECTRONICS MANUFACTURING

DESIGN-TO-DEVICE RAIL → SEMICONDUCTOR BUSINESS-MODEL MATRIX → FRONT-END VERSUS BACK-END → NODE AND LITHOGRAPHY LADDER → ADVANCED INTEGRATION MAP → MATERIALS TAXONOMY

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ● APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

00

STAGE 00 DESIGN-TO-DEVICE RAIL

DESIGN-TO-DEVICE RAIL · ARCHITECTURE, VERIFICATION AND TAPE-OUT · TRANSISTOR AND INTERCONNECT LAYERS ON WAFER · WAFER BECOMES INDIVIDUAL DIES · ASSEMBLE, TEST, MARK AND PACKAGE · DOWNSTREAM ELECTRONICS INTEGRATION

DESIGN / IP / EDA — architecture, verification and tape-out → FOUNDRY / FAB — transistor and interconnect layers on wafer → DICING — wafer becomes individual dies → ATMP / OSAT — assemble, test, mark and package → BOARD / MODULE / DEVICE — downstream electronics integration

STARTING CONDITION / INPUT
- DESIGN / IP / EDA — architecture, verification and tape-out
- FOUNDRY / FAB — transistor and interconnect layers on wafer

SCIENTIFIC OR ENGINEERING MECHANISM
- DICING — wafer becomes individual dies
- ATMP / OSAT — assemble, test, mark and package

OUTPUT / FAILURE BOUNDARY
- BOARD / MODULE / DEVICE — downstream electronics integration
- DESIGN / IP / EDA — architecture, verification and tape-out

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BOARD / MODULE / DEVICE — downstream electronics integration.

01

STAGE 01 SEMICONDUCTOR BUSINESS-MODEL MATRIX

SEMICONDUCTOR BUSINESS-MODEL MATRIX · PHYSICAL WAFER-MANUFACTURING PLANT · DESIGNS AND OUTSOURCES WAFER MANUFACTURE · MANUFACTURES TO CUSTOMERS' DESIGNS · DESIGNS AND FABRICATES ITS OWN PRODUCTS · PLANT TYPE AND FIRM MODEL ANSWER DIFFERENT QUESTIONS

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
FAB — physical wafer-manufacturing plant	FOUNDRY — manufactures to customers' designs	RULE — plant type and firm model answer different questions
FABLESS — designs and outsources wafer manufacture	IDM — designs and fabricates its own products	FAB — physical wafer-manufacturing plant

SYSTEM / DEVICE / INSTITUTION
- FAB — physical wafer-manufacturing plant
- FABLESS — designs and outsources wafer manufacture

DECISIVE TECHNICAL DISTINCTION
- FOUNDRY — manufactures to customers' designs
- IDM — designs and fabricates its own products

STATUS / UNIT / EVIDENCE LIMIT
- RULE — plant type and firm model answer different questions
- FAB — physical wafer-manufacturing plant

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE — plant type and firm model answer different questions.

02

STAGE 02 FRONT-END VERSUS BACK-END

FRONT-END VERSUS BACK-END · FRONT END · DEPOSITION, LITHOGRAPHY, ETCHING, DOPING, METALLISATION · PROCESSED WAFER CONTAINING DIES · BACK END · BONDING, ENCAPSULATION, TESTING, MARKING, PACKAGING · PROCESS DESCRIPTION: OSAT · OUTSOURCED SERVICE MODEL

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
FRONT END — deposition, lithography, etching, doping, metallisation	BACK END — bonding, encapsulation, testing, marking, packaging	TRAP — back-end capacity is not front-end fabrication
OUTPUT — processed wafer containing dies	ATMP — process description; OSAT — outsourced service model	FRONT END — deposition, lithography, etching, doping, metallisation

SYSTEM / DEVICE / INSTITUTION
- FRONT END — deposition, lithography, etching, doping, metallisation
- OUTPUT — processed wafer containing dies

DECISIVE TECHNICAL DISTINCTION
- BACK END — bonding, encapsulation, testing, marking, packaging
- ATMP — process description; OSAT — outsourced service model

STATUS / UNIT / EVIDENCE LIMIT
- TRAP — back-end capacity is not front-end fabrication
- FRONT END — deposition, lithography, etching, doping, metallisation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP — back-end capacity is not front-end fabrication.

03

STAGE 03 NODE AND LITHOGRAPHY LADDER

NODE AND LITHOGRAPHY LADDER · BROAD AUTOMOTIVE, INDUSTRIAL AND POWER USES · MATURE AND MID-RANGE LITHOGRAPHY ROUTE · LEADING EDGE · RISING COMPLEXITY AND CAPITAL INTENSITY · CONCENTRATED EQUIPMENT AND EXPORT-CONTROL CHOKEPOINT · VERIFY EVERY PROJECT-SPECIFIC NODE CLAIM OFFICIALLY

MATURE / MID-RANGE — broad automotive, industrial and power uses

DUV — mature and mid-range lithography route → LEADING EDGE — rising complexity and capital intensity → EUV — concentrated equipment and export-control chokepoint

STARTING CONDITION / INPUT
- MATURE / MID-RANGE → broad automotive, industrial and power uses
- DUV → mature and mid-range lithography route

SCIENTIFIC OR ENGINEERING MECHANISM
- LEADING EDGE → rising complexity and capital intensity
- EUV → concentrated equipment and export-control chokepoint

OUTPUT / FAILURE BOUNDARY
- RULE → verify every project-specific node claim officially
- MATURE / MID-RANGE → broad automotive, industrial and power uses

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE — verify every project-specific node claim officially.

04

STAGE 04 ADVANCED INTEGRATION MAP

ADVANCED INTEGRATION MAP · SPECIALISED DIES COMBINED AS ONE SYSTEM · DENSER PACKAGE-LEVEL INTEGRATION · THROUGH-SILICON VIAS · VERTICAL INTERCONNECT ROUTE · HETEROGENEOUS INTEGRATION · COMBINE UNLIKE FUNCTIONS OR PROCESSES · PERFORMANCE GAINS NEED NOT BEGIN WITH SMALLER LITHOGRAPHY

CHIPLETS — specialised dies combined as one system

2.5D / 3D — denser package-level integration → THROUGH-SILICON VIAS — vertical interconnect route → HETEROGENEOUS INTEGRATION — combine unlike functions or processes

CONCEPT / ARCHITECTURE
- CHIPLETS — specialised dies combined as one system
- 2.5D / 3D — denser package-level integration

MECHANISM / APPLICATION
- THROUGH-SILICON VIAS — vertical interconnect route
- HETEROGENEOUS INTEGRATION — combine unlike functions or processes

READINESS / STATUS LIMIT
- INSIGHT — performance gains need not begin with smaller lithography
- CHIPLETS — specialised dies combined as one system

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INSIGHT — performance gains need not begin with smaller lithography.

05

STAGE 05 MATERIALS TAXONOMY

MATERIALS TAXONOMY · COMPOUND SEMICONDUCTORS · APPLICATION FAMILIES · SILICON PHOTONICS · SILICON OPTICAL-INTERCONNECT PLATFORM · POLICY GROUPING · MAY PLACE UNLIKE TECHNOLOGIES TOGETHER · SCHEME NOMENCLATURE DOES NOT CHANGE MATERIAL PHYSICS

COMPOUND SEMICONDUCTORS — GaN / GaAs / InP / SiC

POWER / RF / OPTOELECTRONICS — application families → SILICON PHOTONICS — silicon optical-interconnect platform → POLICY GROUPING — may place unlike technologies together

CONCEPT / ARCHITECTURE
- COMPOUND SEMICONDUCTORS — GaN / GaAs / InP / SiC
- POWER / RF / OPTOELECTRONICS — application families

MECHANISM / APPLICATION
- SILICON PHOTONICS — silicon optical-interconnect platform
- POLICY GROUPING — may place unlike technologies together

READINESS / STATUS LIMIT
- RULE — scheme nomenclature does not change material physics
- COMPOUND SEMICONDUCTORS — GaN / GaAs / InP / SiC

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE — scheme nomenclature does not change material physics.

06

STAGE 06 FAB VIABILITY CHAIN

FAB VIABILITY CHAIN · RELIABLE POWER + ULTRA-PURE WATER · STABLE PROCESSING · GASES + CHEMICALS + EQUIPMENT · CONTROLLED PROCESS STEPS · REPEATED RUNS · YIELD LEARNING · RELIABILITY EVIDENCE

RELIABLE POWER + ULTRA-PURE WATER — stable processing

GASES + CHEMICALS + EQUIPMENT — controlled process steps → REPEATED RUNS — yield learning → RELIABILITY EVIDENCE — customer qualification

STARTING CONDITION / INPUT
- RELIABLE POWER + ULTRA-PURE WATER → stable processing
- GASES + CHEMICALS + EQUIPMENT → controlled process steps

SCIENTIFIC OR ENGINEERING MECHANISM
- REPEATED RUNS → yield learning
- RELIABILITY EVIDENCE → customer qualification

OUTPUT / FAILURE BOUNDARY
- OUTCOME → inauguration alone is not competitive production
- RELIABLE POWER + ULTRA-PURE WATER → stable processing

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OUTCOME — inauguration alone is not competitive production.

07

STAGE 07 INSTITUTION AND SCHEME MAP

INSTITUTION AND SCHEME MAP · PARENT MINISTRY · IMPLEMENTING AGENCY · SEMICON INDIA · SCHEME PACKAGE · DESIGN-FOCUSED CHANNEL · SEPARATE COMPLEMENTARY ELECTRONICS INCENTIVES

MeitY — parent ministry

ISM — implementing agency → SEMICON INDIA — scheme package → DLI — design-focused channel

CONCEPT / ARCHITECTURE
- MeitY → parent ministry
- ISM → implementing agency

MECHANISM / APPLICATION
- SEMICON INDIA → scheme package
- DLI → design-focused channel

READINESS / STATUS LIMIT
- SPECS / PLI → separate complementary electronics incentives
- MeitY → parent ministry

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SPECS / PLI — separate complementary electronics incentives.

08

STAGE 08 SEMICON 1.0 TO 2.0

SEMICON 1.0 TO 2.0 · SEMICON INDIA 1.0 · RS 76,000 CRORE UMBRELLA · FOUR CHANNELS · FABS, DISPLAYS, COMPOUND/SENSOR/ATMP, DLI · 15 JUL 2026 · SEMICON 2.0 CABINET APPROVAL · SEMICON 2.0

SEMICON INDIA 1.0 — Rs 76,000 crore umbrella → FOUR CHANNELS — fabs, displays, compound/sensor/ATMP, DLI → 15 JUL 2026 — Semicon 2.0 Cabinet approval → SEMICON 2.0 — Rs 1,27,500 crore approved outlay → SIX PILLARS — design, tools/materials, fabs, back end, R&D, talent

CONCEPT / ARCHITECTURE
- SEMICON INDIA 1.0 → Rs 76,000 crore umbrella
- FOUR CHANNELS → fabs, displays, compound/sensor/ATMP, DLI

MECHANISM / APPLICATION
- 15 JUL 2026 — Semicon 2.0 Cabinet approval
- SEMICON 2.0 → Rs 1,27,500 crore approved outlay

READINESS / STATUS LIMIT
- SIX PILLARS → design, tools/materials, fabs, back end, R&D, talent
- SEMICON INDIA 1.0 → Rs 76,000 crore umbrella

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SIX PILLARS — design, tools/materials, fabs, back end, R&D, talent.

09

STAGE 09 FACILITY EVIDENCE MAP

FACILITY EVIDENCE MAP · FAB: LATER NAMED COMPOUND-FAB PLUS ATMP APPROVAL · OFFICIALLY IDENTIFIED OSAT · ATMP LOCATION · OFFICIALLY IDENTIFIED BACK-END CLUSTER AND INAUGURATIONS · SUCHI SEMICON OSAT APPROVAL · LOCATION MUST TRAVEL WITH PROJECT AND STATUS VERB

DHOLERA — fab; later named compound-fab plus ATMP approval

MORIGAON — officially identified OSAT / ATMP location → SANAND — officially identified back-end cluster and inaugurations → SURAT — Suchi Semicon OSAT approval

CONCEPT / ARCHITECTURE
- DHOLERA → fab; later named compound-fab plus ATMP approval
- MORIGAON → officially identified OSAT / ATMP location

MECHANISM / APPLICATION
- SANAND → officially identified back-end cluster and inaugurations
- SURAT → Suchi Semicon OSAT approval

READINESS / STATUS LIMIT
- RULE → location must travel with project and status verb
- DHOLERA → fab; later named compound-fab plus ATMP approval

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE — location must travel with project and status verb.

10

STAGE 10 STATUS FIREWALL

STATUS FIREWALL · POLICY PERMISSION AND SUPPORT DECISION · FINANCING OR CONSTRUCTION MILESTONE · PHYSICAL OR DEMONSTRATION MILESTONE · QUALIFIED COMMERCIAL SUPPLY · SEPARATE OPERATING EVIDENCE · NEVER SKIP A RUNG OR INVENT PRODUCTION

APPROVED — policy permission and support decision

AGREEMENT / GROUNDBREAKING — financing or construction milestone → INAUGURATED / PRESENTED — physical or demonstration milestone → QUALIFIED COMMERCIAL SUPPLY — separate operating evidence

SYSTEM / DEVICE / INSTITUTION
- APPROVED → policy permission and support decision
- AGREEMENT / GROUNDBREAKING → financing or construction milestone

DECISIVE TECHNICAL DISTINCTION
- INAUGURATED / PRESENTED → physical or demonstration milestone
- QUALIFIED COMMERCIAL SUPPLY → separate operating evidence

STATUS / UNIT / EVIDENCE LIMIT
- RULE → never skip a rung or invent production
- APPROVED → policy permission and support decision

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE — never skip a rung or invent production.

11

STAGE 11 PYQ ANSWER SPINE

PYQ ANSWER SPINE · VALUE CHAIN, FAB MODELS AND ATMP · ISM, FOUR SEMICON 1.0 CHANNELS AND SEMICON 2.0 PILLARS · TOOLS, UTILITIES, TALENT, YIELD AND QUALIFICATION · DHRUV64 AND FACILITIES WITH EXACT STATUS VERBS · NO PROJECT TOTAL, NODE, COMMERCIAL OUTPUT OR ANSWER KEY

DEFINE — value chain, fab models and ATMP / OSAT → MAP — ISM, four Semicon 1.0 channels and Semicon 2.0 pillars → ANALYSE — tools, utilities, talent, yield and qualification → EVIDENCE — DHRUV64 and facilities with exact status verbs → QUALIFY — no project total, node, commercial output or answer key

CONCEPT / ARCHITECTURE
- DEFINE — value chain, fab models and ATMP / OSAT
- MAP — ISM, four Semicon 1.0 channels and Semicon 2.0 pillars

MECHANISM / APPLICATION
- ANALYSE — tools, utilities, talent, yield and qualification
- EVIDENCE — DHRUV64 and facilities with exact status verbs

READINESS / STATUS LIMIT
- QUALIFY — no project total, node, commercial output or answer key
- DEFINE — value chain, fab models and ATMP / OSAT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ANALYSE — tools, utilities, talent, yield and qualification.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · PACKAGING-FIRST VERSUS FAB-FIRST DEBATE · INDIA MAY OBTAIN FASTER LEARNING, CUSTOMER RELATIONSHIPS AND · EXPORT INTEGRATION THROUGH ATMP/OSAT AND COMPOUND-SEMICONDUCTOR NICHES BEFORE MASTERING · SUBSIDY VERSUS ECOSYSTEM DEBATE · GENEROUS FISCAL SUPPORT CAN ATTRACT PROPOSALS, BUT · PROCESS ENGINEERS, UTILITIES, MATERIALS AND ANCHOR CUSTOMERS DETERMINE WHETTHER · STRATEGIC AUTONOMY VERSUS COST EFFICIENCY

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH
- CAUTION: Packaging-first versus fab-first debate: India may obtain faster learning, customer relationships and
- export integration through ATMP/OSAT and compound-semiconductor niches before mastering complex leading-edge fabrication.
- CAUTION: Subsidy versus ecosystem debate: generous fiscal support can attract proposals, but
- process engineers, utilities, materials and anchor customers determine whether plants remain viable.

READINESS, UNIT OR STATUS LIMIT
- CAUTION: Strategic autonomy versus cost efficiency: semiconductors are treated as strategic infrastructure, yet full autarky is unrealistic
- the more realistic goal is selective resilience in critical layers.
- CAUTION: Technology-neutral versus technology-specific support: India must decide whether to prioritize broad ecosystem depth or
- a few strategically valuable segments such as power electronics, sensors, compound semiconductors and packaging.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE
- CAUTION: Semiconductors increasingly shape AI, telecom, defence electronics, automobiles, power systems and industrial automation.
- CAUTION: Domestic packaging and design capability can reduce vulnerability even when India still imports large volumes of wafers or equipment.
- CAUTION: Semiconductor success improves the quality of India's electronics manufacturing
- story by moving beyond screwdriver assembly toward deeper value capture.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels